AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 29 · STUDENT LESSON

Feature Engineering

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Name the formula. Match the units. Freeze the training state.

Construct legitimate features while keeping arithmetic,availability and predictive usefulness separate.

Keep one habit: Complete geometry contract → pictorial centering → zero/missing/mixed-unit probes → controlled saved-result comparison → independent new area.

MAIN LESSON · SECTION 1

Which information can this feature legitimately use?

> A useful formula needs units,availability and a missing-value policy.

Learning goals

I can create features available at **prediction** time.

I can fit feature **transformation**s using **training evidence** only.

I can compare a feature change without changing the evaluation set.

Recall training-only preparation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.
- Why must pre**process**ing be fitted on **training** only?

Readiness for the association extension

- Can a survey of only one club establish the preference of all students?

> **My readiness explanation**

End goal

By the end, I can explain **feature engineering** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Feature engineering transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.



LEARNING TARGET Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Relationships and confounding

This week's end goal

MAIN LESSON · SECTION 2

Construct only from information available at prediction time

> A future outcome is not a legitimate input just because it correlates with the target.

Main geometric feature and fictional scope

A useful feature represents information available when the **prediction** is made. In a fictional parcel task, length and width are measured before packing, so area=length*width is available.

The final delivery delay is not available when predicting dispatch time. A strong correlation cannot make a future value legitimate.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **feature engineering**.

▼ **Optional extension · Complete practice/score association**

In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an **association**. Prior experience could affect both practice choices and scores.

The table alone cannot establish the effect of adding practice time.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **feature engineering** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Relationships and confounding

MAIN LESSON · SECTION 3

Keep construction and training-fitted preparation distinct

> Area is a declared formula;centering learns one value from training.

Complete dimensions,formulas and unit contract

Feature step	Complete main dimensions / rule	Unit or boundary
Training rows	T1 length2, width 3; T2 length4, width 2; T3 length6, width 1, both dimensionscm	Areas6, 8, 6cm²
Fit centering state	Training mean area(6+8+6)/3=20/3	cm²; learned from training only
Validation V1	Length5 cm, width 2 cm→area 10cm²	Centered area 10-20/3=10/3cm²
Separate ratio example	Length12 cm, width 30 mm=3 cm→length_over_width4	Dimensionless ratio with declared order
Zero/missing width	Area formula permits measured 0 width→area 0; length/width cannot divide by0. Missing width is unobserved.	Return ratio unavailable for 0; request/reject missing dimensions under this contract.

Training rectangles have lengths [2, 4, 6] and widths [3, 2, 1], so areas are[6, 8, 6] square cm. **Training** mean area is 20/3. A validation rectangle length 5 width 2 has area 10 and centered area 10-20/3=10/3.

Freeze the **training** mean; do not recompute it with validation.

Explain the mechanism from its evidence

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

▼ Optional extension · Printed association records and confounding explanation

```
rows = [{"hours": 1, "score": 4},
        {"hours": 2, "score": 6},
        {"hours": 3, "score": 8}]
for row in rows:
    print(row["hours"], row["score"])
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Observed	More practice with higher scores	Association
Possible other factor	Prior experience	May affect both
Missing evidence	Comparable starting conditions	Limits causal claim

Read each row from left to right and connect the evidence to the stated rule.

The numbers rise together in these three invented records. A description such as higher practice hours went with higher scores is supported. A claim that one extra hour guarantees two more points goes beyond the **observation**s.

No statistical calculation is required to see this limit.

The three records show that higher practice time goes with higher scores. They do not tell us what would happen if the same learner practised for one more hour.

Prior experience could make someone both practise more and score higher. That is a possible **confounder**: another influence connected to both quantities. First describe the pattern accurately.

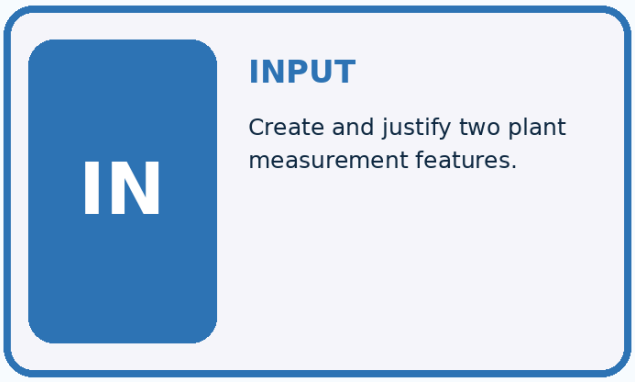
Then propose a **comparison** that addresses alternative explanations rather than treating the pattern as a guarantee.

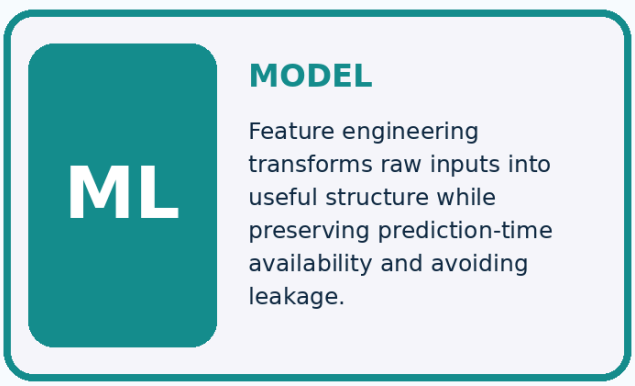
▼ Earlier wording, alternatives and picture transcripts

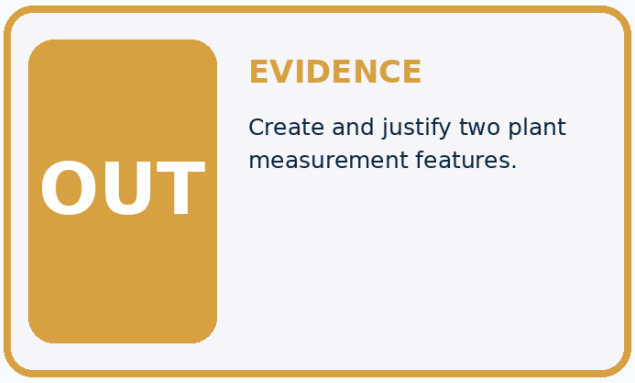
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Create and justify two plant measurement features.

MODEL IDEA **Feature engineering** transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Raw leaf length=8 cm and width=4 cm. Derived aspect_ratio = 4 ÷ 8 = 0.5.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Relationships and confounding

Separate a pattern from a cause

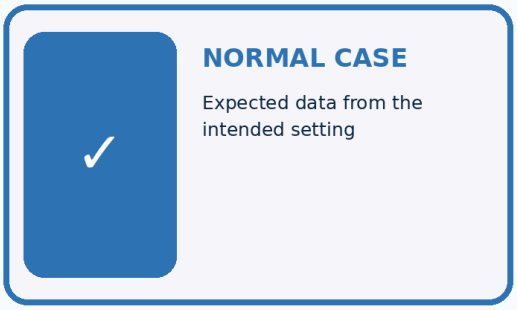
Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

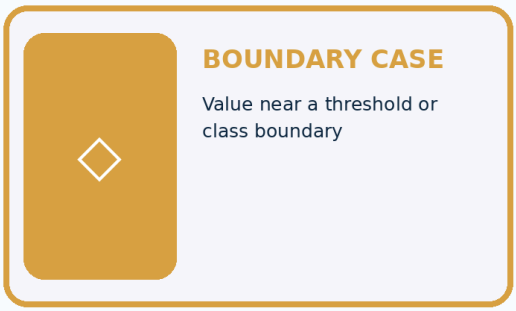
Compare available measurements with leaked or missing information

> An engineered number can be valid arithmetic without useful predictive evidence.

Four evaluation conditions



NORMAL Expected data and expected behavior.

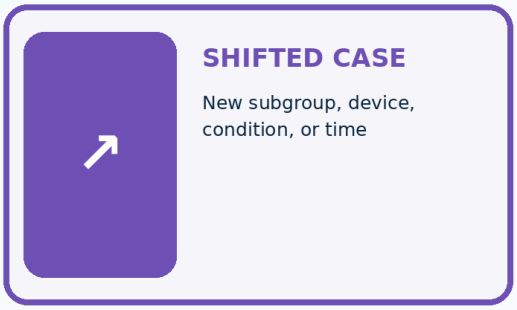


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Include the query in the centering mean.

▼ **Check after predicting**

Prediction: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

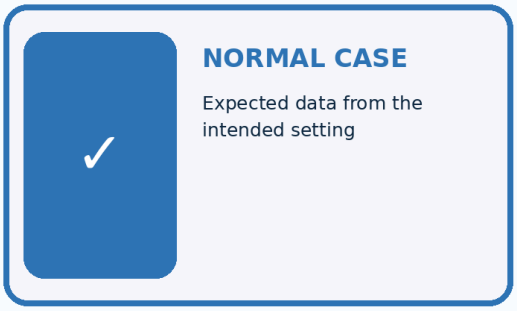
Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

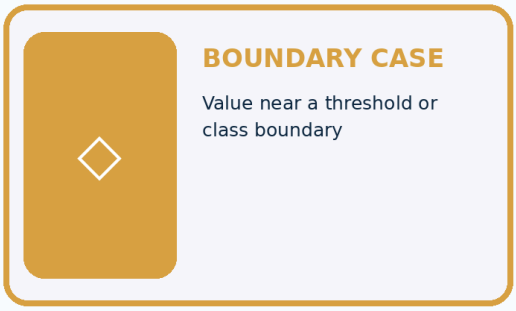
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

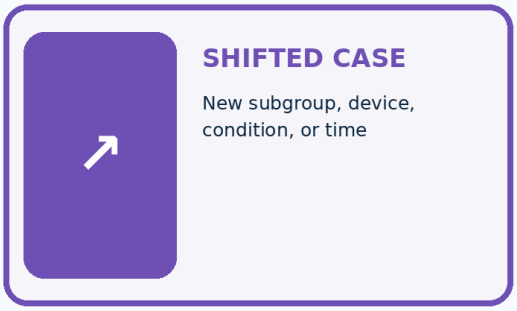
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the frozen training mean beside the query formula

> The query should not quietly redefine the fitted transformation.

Complete pictured facts and separate-fixture scope

Engineer a feature with a **transformation** defined before evaluating a new leaf.

Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

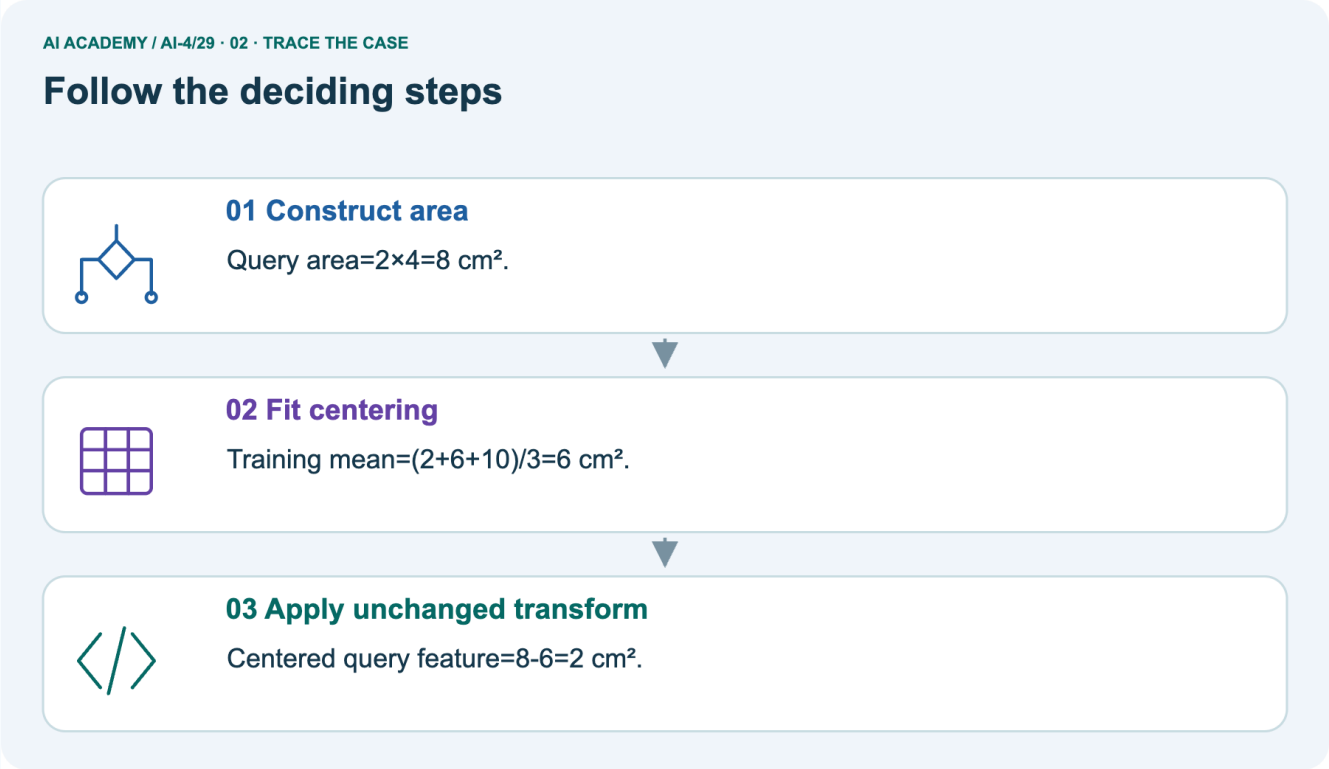
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main training areas 6, 8, 6cm²give mean 20/3 and V1 area10 centered10/3. Picture is separate:training areas 2, 6, 10 give mean 6cm²; query 2×4=8cm²centers to2cm². Including that query in its mean gives(2+6+10+8)/4=6.5cm²and query 1.5cm², showing evaluation

information changed fitted preprocessing. Centering preserves square-area units; it is not a unitless ratio.

These rectangle/bounding-card examples do not establish actual leaf area, disease diagnosis or feature benefit.

Trace area,centering and unchanged state



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Construct area	Query area=2×4=8 cm².
2. Fit centering	Training mean=(2+6+10)/3=6 cm².
3. Apply unchanged transform	Centered query feature=8-6=2 cm².

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

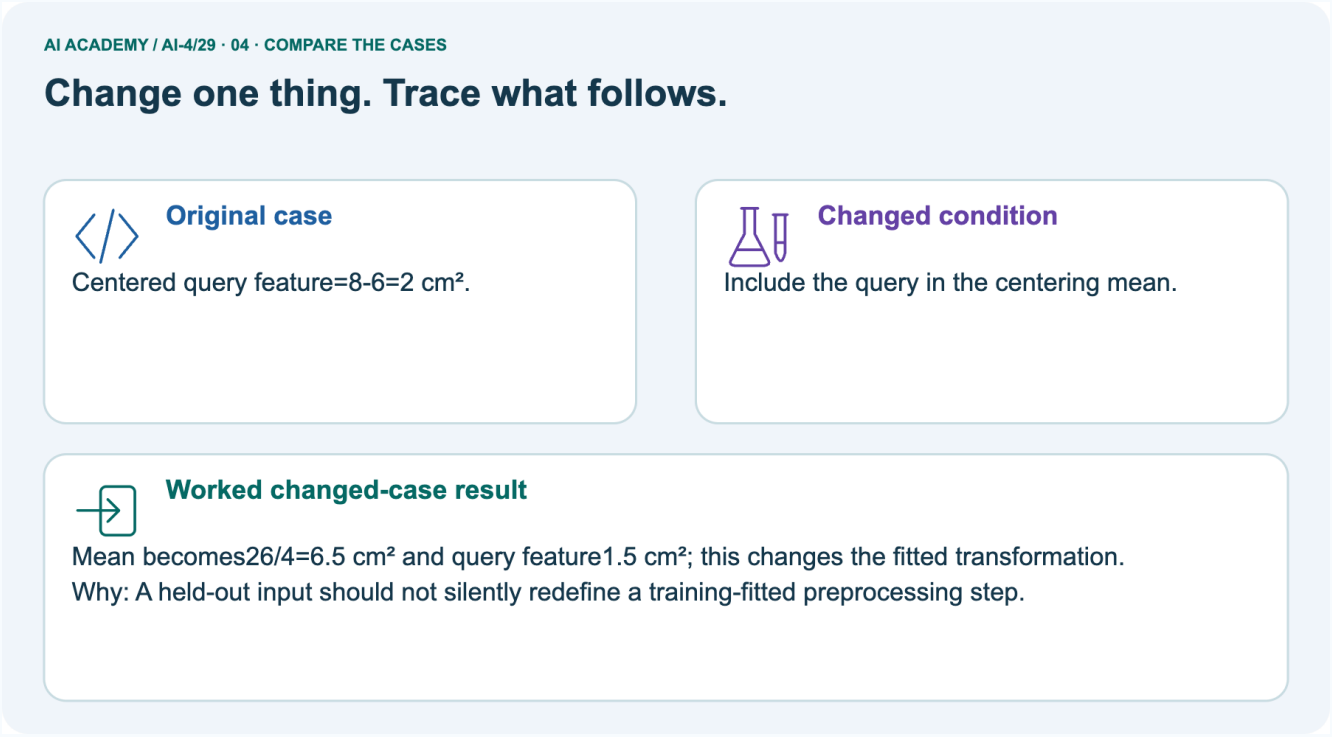
Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict including the query in the mean

- Change one thing: Include the query in the centering mean. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain measurement and usefulness limits

- Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

- Read the labelled picture: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to main V1centering

Feature engineering at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Raw measurements	Length and width	Available before prediction
Derived area	Multiply length by width	Units square cm
Centering	Subtract training mean	Reuse frozen value

Area is a deterministic transformation; it adds a representation, not a new measured fact. It may help a limited model express a relationship, or may do nothing useful. Compare original versus extended features under the same split, metric and development policy.

Center the validation area using the frozen training mean. Explain the steps or evidence behind your response.

▼ Optional confounder question

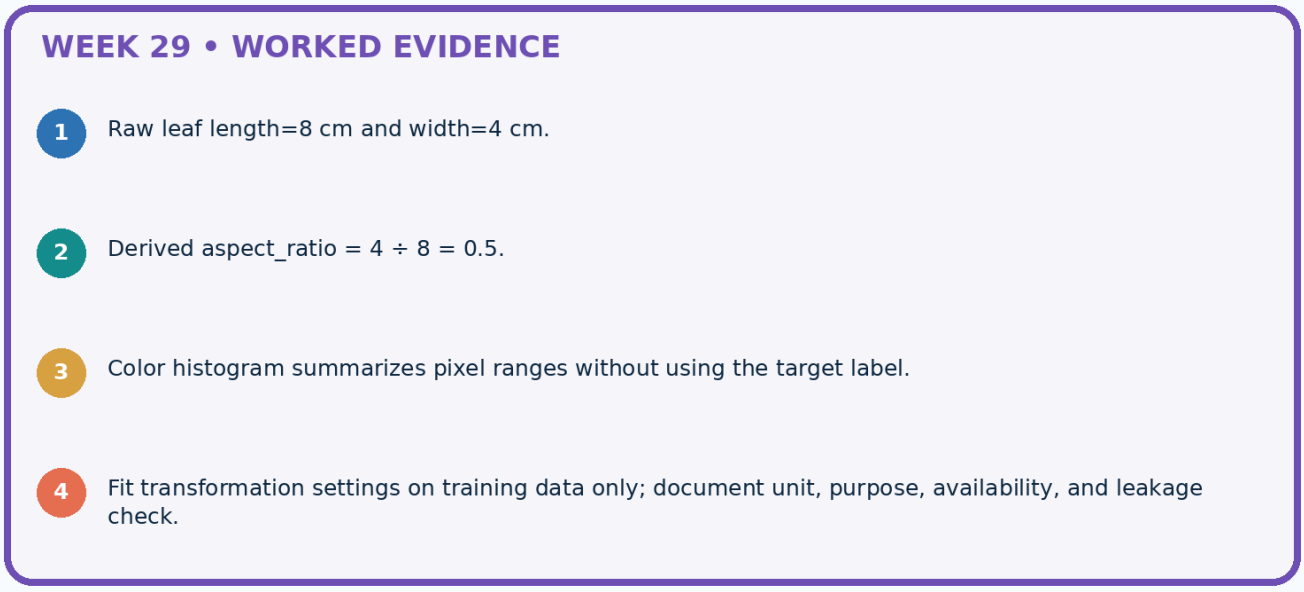
Name one factor besides practice duration that could influence both time chosen and score.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.



STEP 1 Raw leaf length=8 cm and width=4 cm.

STEP 2 Derived aspect_ratio = 4 ÷ 8 = 0.5.

STEP 3 Color histogram summarizes pixel ranges without using the target label.

STEP 4 Fit **transformation** settings on **training** data only; document unit, purpose, **availability**, and **leakage** check.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

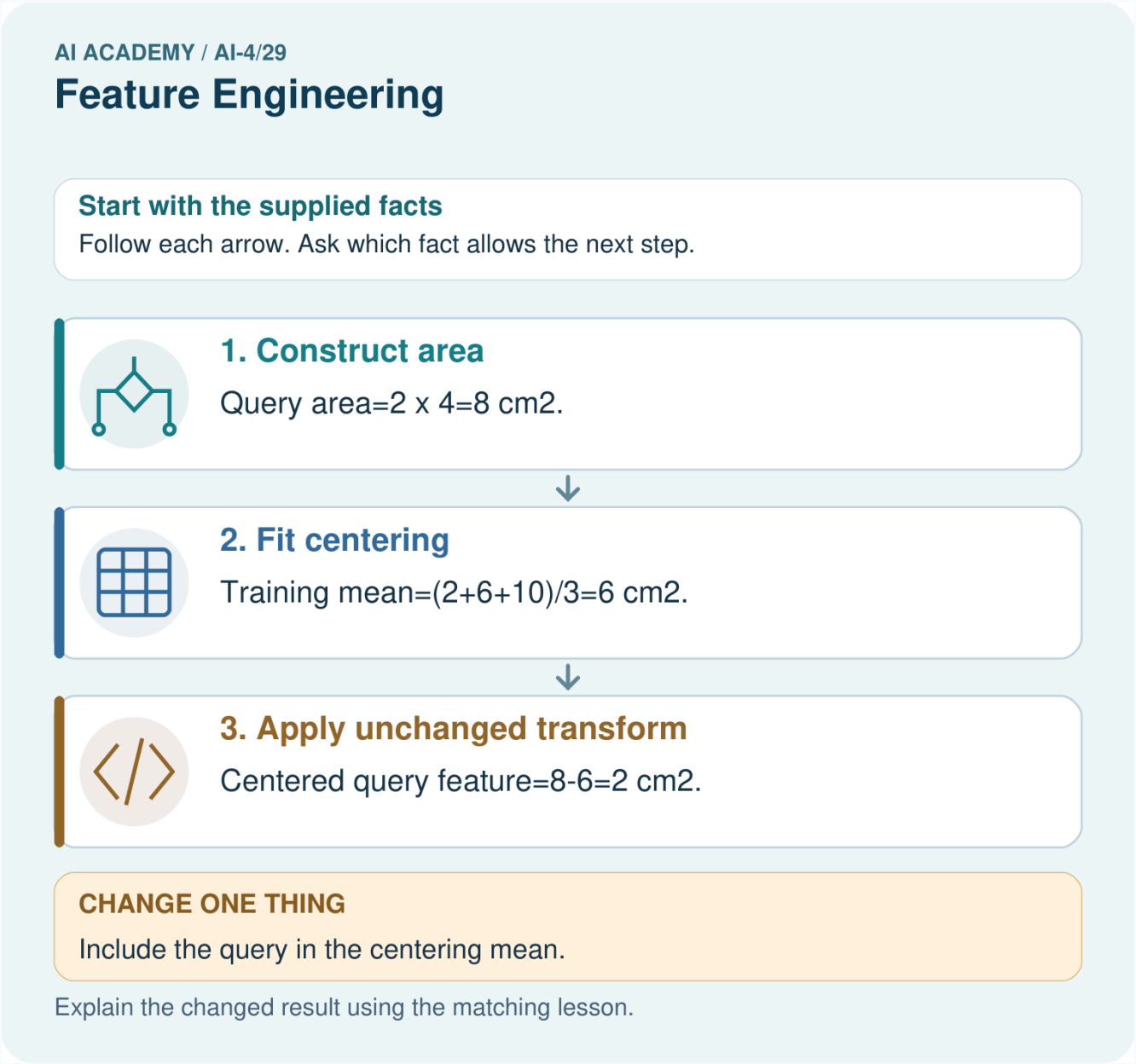
Apply the connected idea: Relationships and confounding

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Feature Engineering

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a plausible feature used as proof of benefit or cause

> Check the source,time and uncertainty before trusting the number.

Diagnose the main claim

- Diagnose one **cause** and name the cleanest next experiment:

CLAIM TO CHALLENGE Creating many features automatically gives the model more useful information.

Explain main feature limits

- An identifier may appear predictive be**cause** related objects leak across splits. Inspect grouped records and proxy risks before adding it.

  Never include private student details for this exercise; use the supplied fictional measurements.

> **How would you test adding area without claiming it must help?**

▼ Optional paper-only association comparison and limits

A fairer **comparison** could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions.

In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Even a better comparison has only four fictional groups. What should be reported?

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Creating many features automatically gives the model more useful information. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

Apply the connected idea: Relationships and confounding

MAIN LESSON · SECTION 8

Practise a reproducible feature log with less help

> A missing observation should remain different from a measured zero.

Try supported V1transformation

Use workbook W2 for the connected example “**Feature engineering**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Record source,formula,units,time and missing policy

Build a feature log with name, units, source, time available, **transformation** and missing-value policy. If width is missing, follow a declared **training**-derived imputation or return a clear unavailable result.

Do not silently use zero, which describes a zero-area object.

Explain each transformation from evidence

- Use the worked case for **Feature engineering**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the saved state needed by the picture

- What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Reserve the new independent area/mean task

This is supported practice. Keep the separately named independent case for an attempt before feedback.

<details>
<summary>Optional association wording and practice</summary>

A chart title Practice **cause**s success overstates the **evidence**. Replace it with Practice time and scores in three fictional records, and describe the possible **confounder**.

Better wording does not repair the study, but it makes the report honest.

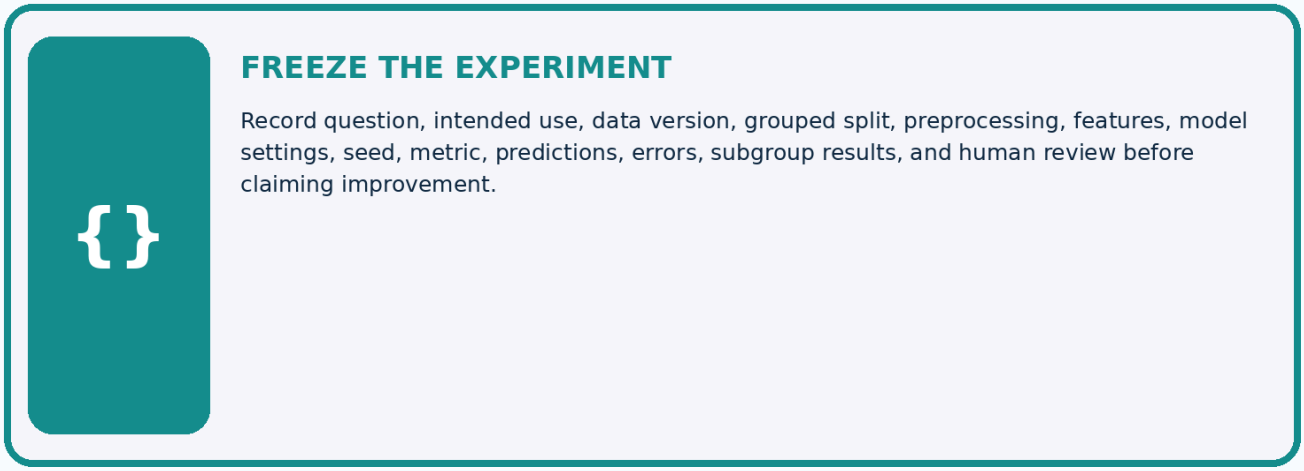
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion
</details>

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + **leakage** check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: Relationships and confounding

Practise with less help: Transform validation

Check your reasoning

Practise the connected topic: Relationships and confounding

Explain the picture

MAIN LESSON · SECTION 9

Run complete feature construction and a bounded controlled comparison

> All dimensions,unit policies and comparison predictions needed for the investigation are supplied.

Normal,zero,missing,mixed-unit and fitted-state probes

Complete feature probe	Expected bounded check
Normal validation length 5, width 2 cm	Area10cm², centered 10/3 using main training mean 20/3.
Measured boundary length 5, width 0 cm	Area0 under declared arithmetic contract; length/width unavailable. Do not divide by zero.
Missing width None or unknown unit ticks	No justified observed feature; request/reject under this program's contract, not invented 0or assumedcm.
Mixed units length 12 cm, width 30 mm	Convert width to3 cm; length_over_width4. Raw12/30=0.4 compares inconsistent units.

Complete feature probe	Expected bounded check
Evaluation query changes fitted mean	Leakage of held-out features into learned preparation; restore training-only state and document exposure.

Construct,audit and compare

1. Feature Workshop: define rectangle area and explicitly named length_over_width, compute the complete main examples, and record their units and missing/zero policy. Audit a future delivery delay as unavailable at the earlier prediction moment. Predict the optional program output and compare fixed training-state reuse.

   Then inspect every row of the separate C1–C4 controlled prediction comparison before claiming an improvement. A formula's correctness and a feature's utility are separate evidence questions. Keep the selected new training-area/query task for an uncoached first attempt.

Complete optional feature/centering program

```
import math
from statistics import mean


def features(length_cm, width, width_unit='cm'):
    if width_unit not in ('cm', 'mm'):
        raise ValueError('Need an explicit supported width unit')
    if any(type(v) not in (int, float) or not math.isfinite(v) or v < 0
           for v in (length_cm, width)):
        raise ValueError('Need observed finite nonnegative dimensions')
    width_cm = width if width_unit == 'cm' else width / 10
    area = length_cm * width_cm
    ratio = length_cm / width_cm if width_cm != 0 else None
    return {'width_cm': width_cm, 'area_cm2': area, 'length_over_width': ratio}


def fit_area_mean(training):
    if not training:
        raise ValueError('Need nonempty training dimensions')
    return mean(features(length, width)['area_cm2'] for length, width in training)
```

```
training = [(2, 3), (4, 2), (6, 1)]  # Both dimensions cm.
training_mean = fit_area_mean(training)
query = features(5, 2)
print(training_mean, query['area_cm2'], query['area_cm2'] - training_mean)
print(features(12, 30, 'mm'))
print(features(5, 0))  # Zero area is computed; the ratio is unavailable.
# A missing observed dimension is not a measured zero; this contract rejects it.
# No label, future outcome or validation feature enters the fitted training mean.
```

Complete separate C1–C4comparison

Separate fictional leaf-card comparison ID · Length/width cm · Reviewed toy reference · Length-only candidate prediction · Length+area candidate prediction

> **Separate fictional leaf-card comparison ID:** C1
>
> **Length/width cm:** 5 /2
>
> **Reviewed toy reference:** spotted
>
> **Length-only candidate prediction:** spotted
>
> **Length+area candidate prediction:** spotted

> **Separate fictional leaf-card comparison ID:** C2
>
> **Length/width cm:** 3 /3
>
> **Reviewed toy reference:** healthy
>
> **Length-only candidate prediction:** healthy
>
> **Length+area candidate prediction:** spotted

> **Separate fictional leaf-card comparison ID:** C3
>
> **Length/width cm:** 1 /4
>
> **Reviewed toy reference:** spotted
>
> **Length-only candidate prediction:** healthy
>
> **Length+area candidate prediction:** spotted

> **Separate fictional leaf-card comparison ID:** C4
>
> **Length/width cm:** 4 /1
>
> **Reviewed toy reference:** healthy
>
> **Length-only candidate prediction:** healthy
>
> **Length+area candidate prediction:** healthy

Protocol,results and missing-benefit limits

The controlled comparison supplies predictions from two declared pipeline versions with the same training/evaluation sources, learner/settings and review rule, changing only the feature list from length to length+rectangle area. Both are assumed refitted on training only. This is a fictional saved-result fixture, not code that fits those classifiers.

Both score 3/4:the length-only version missesC3; the area version falsely flagsC2. Area did not improve total accuracy on these records. Different errors may matter, but no real consequences or general benefit are established.

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Feature Workshop: design two transformations, compute examples, audit leakage, and compare one controlled validation result.

1. Write the hypothesis and expected **evidence** before running.

2. Feature Workshop: design two **transformation**s, compute examples, audit **leakage**, and compare one controlled validation result.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: construct and center a new area feature

> Keep training state,query arithmetic and square units separate.

Complete independent inputs and scope

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

Use the selected training areas 4, 8, 12cm²and new rectangle length 3 cm, width 4 cm. The average is fitted from training areas only; then apply it unchanged to the new area. This is a separate area/centering task, not the main 20/3 mean or pictured 6 mean. No prediction labels or model-utility result are supplied for this independent calculation.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 29 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project feature log tied to its evidence

> Another learner should reconstruct the same number without guessing a formula or unit.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

Keep each feature's name and full formula, source fields, measurement meaning, units, time available, missing/zero policy and training-fitted parameters. State a proposed benefit as a hypothesis and compare pipeline versions on the same validation cases with other planned factors fixed.

Preserve leakage/source questions and all old/new errors; do not claim a geometric feature is a measured biological area or a health diagnosis.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish with the formula,state,unit and unanswered benefit question

> Explain why legitimate input information still needs predictive evaluation.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Create features available at **prediction** time

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
transformation	rule converting raw values	
derived feature	new input calculated from existing inputs	
aspect ratio	width divided by length	
feature pipeline	fixed ordered transformations	

Closer-explanation terms

Word	Meaning
Feature engineering	Constructing model inputs from available measurements
Derived feature	An input computed from other inputs
Leakage	Information entering a model that would not be available legitimately

Word	Meaning
Availability	Whether a value exists when a prediction is needed

▼ Extension vocabulary

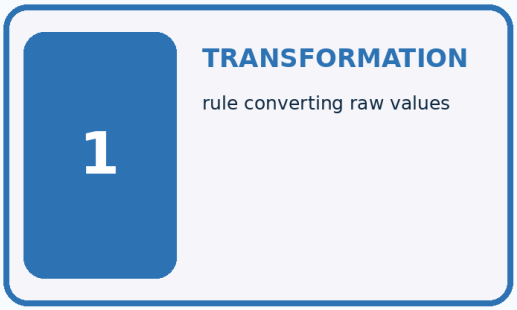
Term	Meaning
Association	Two variables vary together in observed data
Cause	A change in one factor produces a change in another
Confounder	A factor related to both the proposed cause and outcome
Comparison	Examining outcomes under different conditions

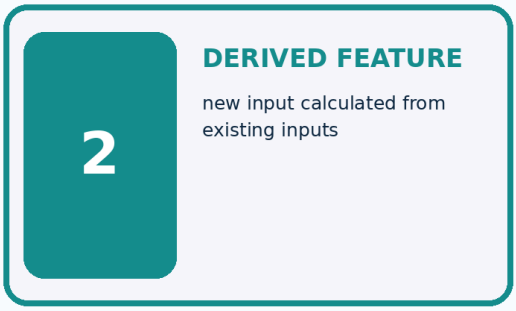
▼ Earlier wording, alternatives and picture transcripts

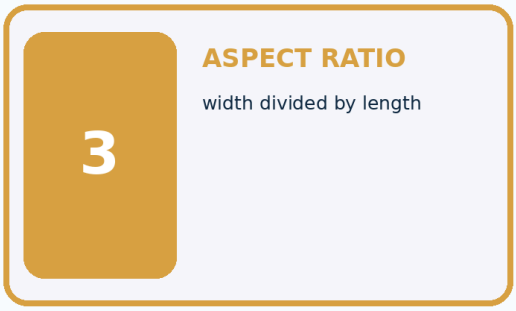
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Creating many features automatically gives the model more useful information.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Relationships and confounding

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **transformation**, **derived feature**, **aspect ratio**, **feature pipeline**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Creating many features automatically gives the model more useful information.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one **cause** and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

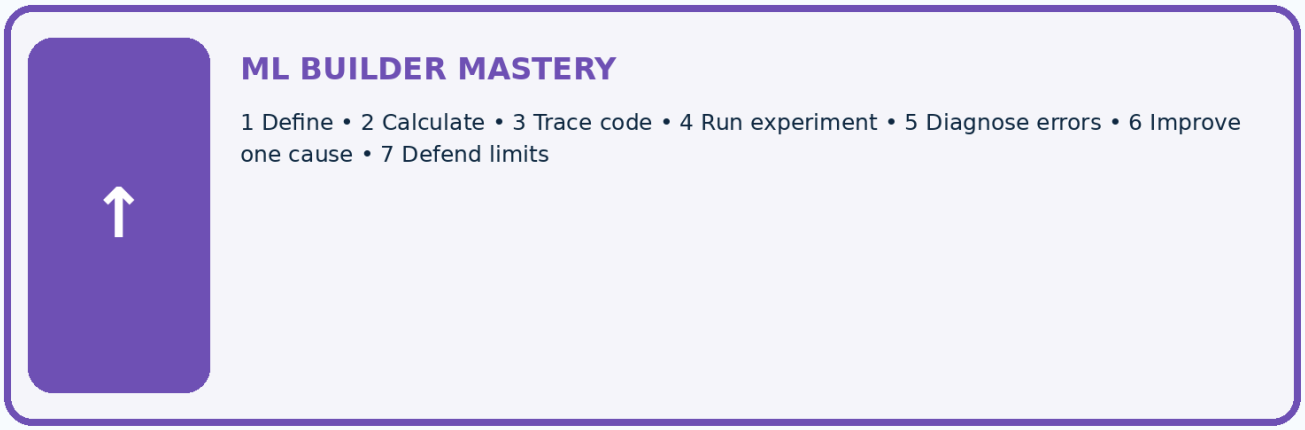
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh training-area4/8/12and3-by-4query task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

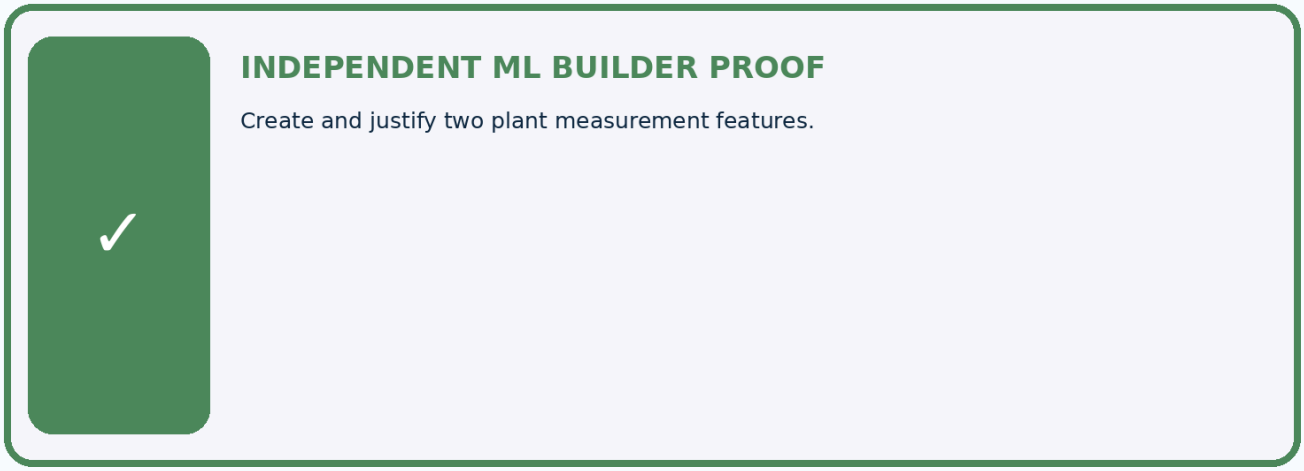
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Create and justify two plant measurement features.

NEXT BRIDGE Carry the experiment **evidence** into: Improve Generalization.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Features are leaf length and width in cm; add length/width.

Compute for8 and2; state boundary checks.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Evidence or stage	Value or result
Length	8 cm
Width	2 cm
Length divided by width	4, unitless
Width zero or missing	Do not divide, handle explicitly

▼ Earlier wording, alternatives and picture transcripts

Week 29 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Engineer a ratio with boundary checks

Features are leaf length and width in cm; add length/width.

Compute for8 and2; state boundary checks.

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Create features available at prediction time	Explain the deciding step, show a trace or test, and state one limit.
Fit feature transformations using training evidence only	Explain the deciding step, show a trace or test, and state one limit.
Compare a feature change without changing the evaluation set	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Engineer a feature with a **transformation** defined before evaluating a new leaf.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Engineer a feature with a **transformation** defined before evaluating a new leaf.

- Which labelled step matches this detail? Centered query feature=8-6=2 cm².
- 1. Construct area
- 2. Fit centering
- 3. Apply unchanged transform
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Apply unchanged transform Centered query feature=8-6=2 cm². Read the picture in order. Centered query feature=8-6=2 cm².

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Include the query in the centering mean. Which conclusion is supported by the supplied facts?
- 1. Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Repair and keep the record

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What **comparison** is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

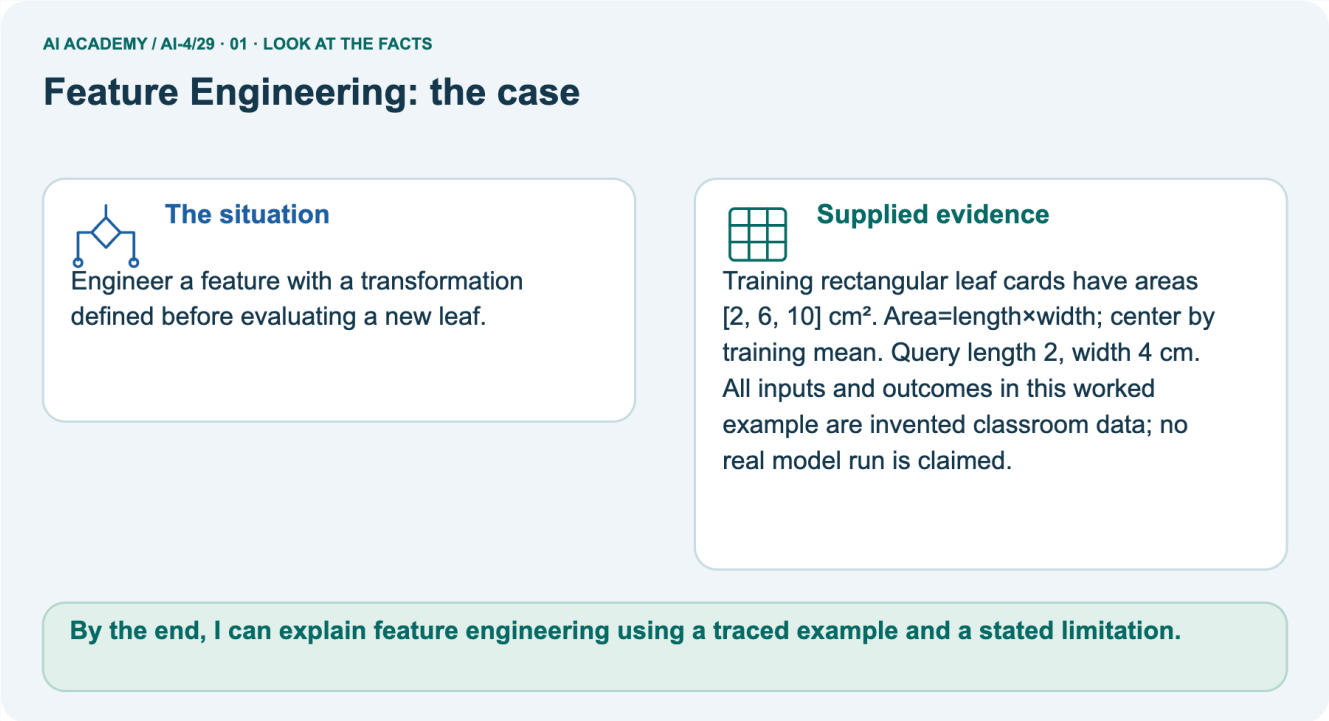
▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Feature Engineering: the case



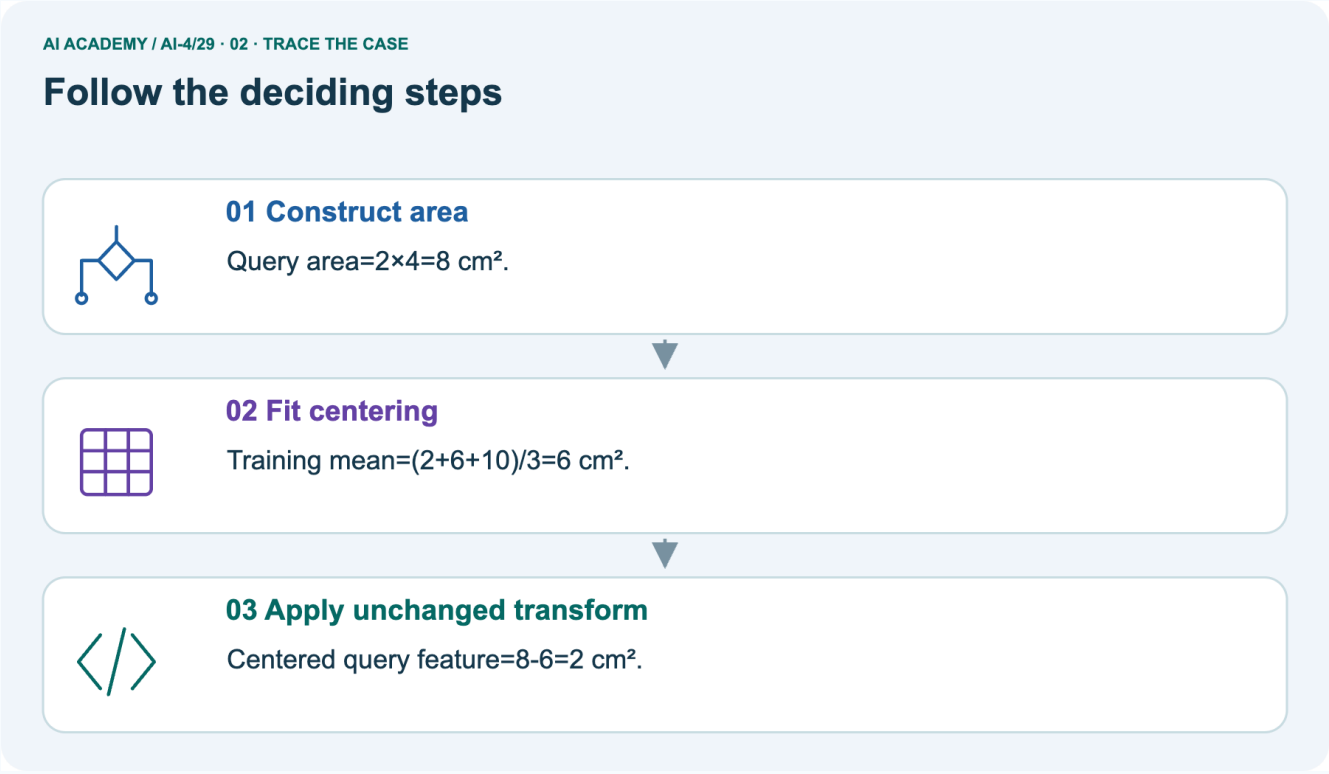
Follow the labelled picture with the matching explanation. Open the vector diagram

Engineer a feature with a **transformation** defined before evaluating a new leaf. Supplied facts: **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

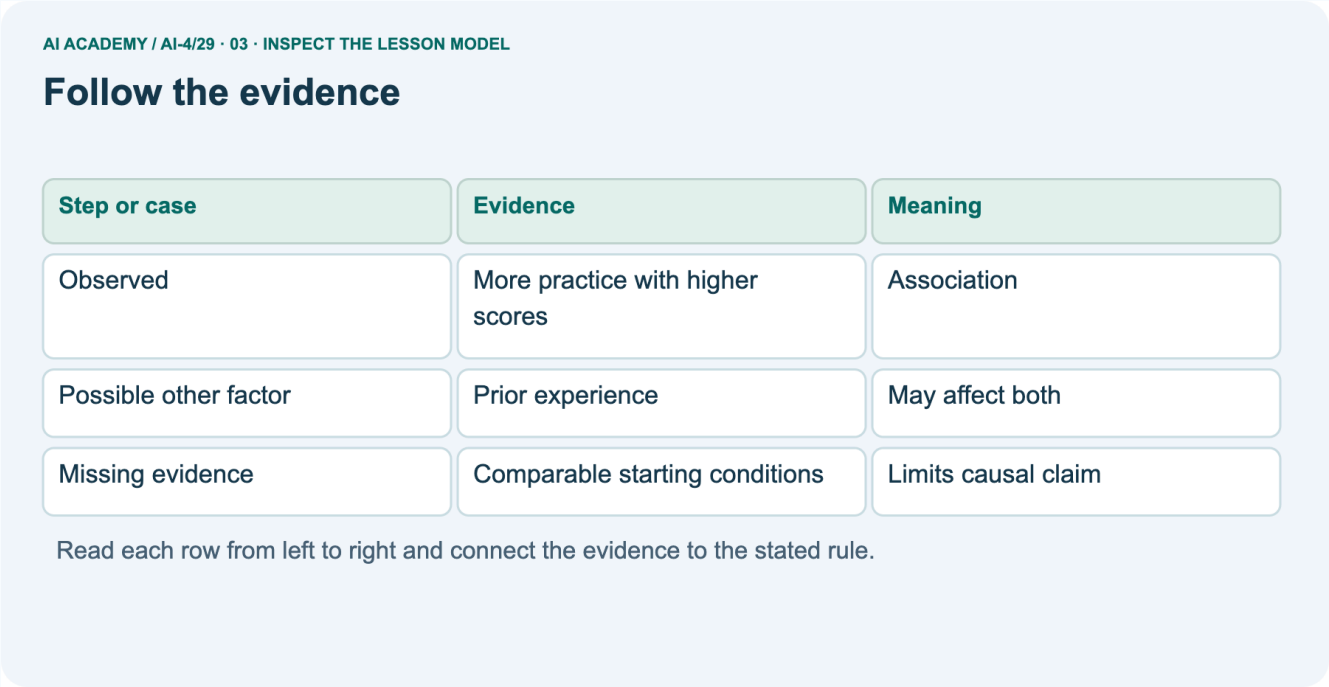


Follow the labelled picture with the matching explanation. Open the vector diagram

Construct area: Query area=2×4=8 cm². → Fit centering: **Training** mean=(2+6+10)/3=6 cm².
→ Apply unchanged transform: Centered query feature=8-6=2 cm².

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

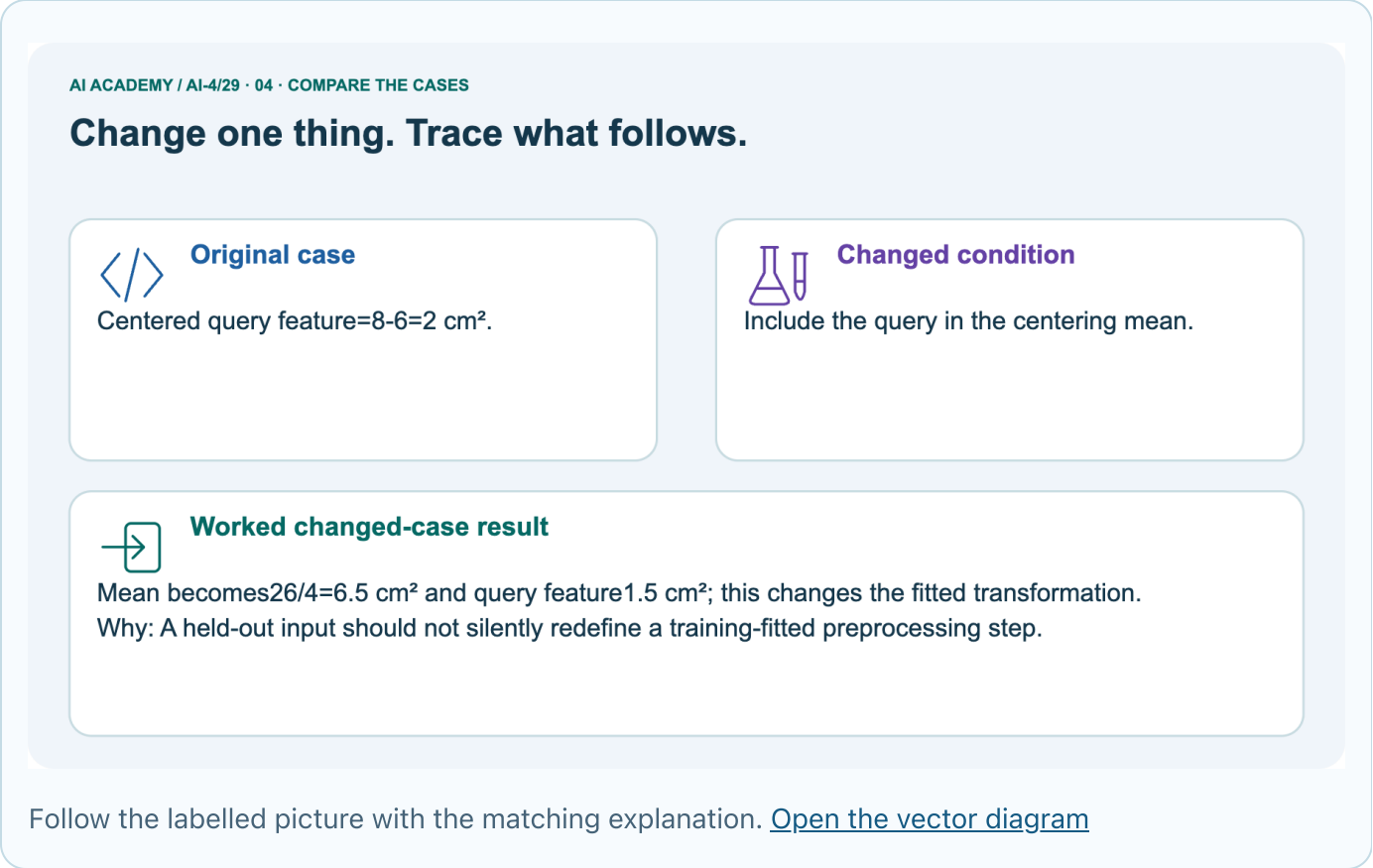


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Observed | More practice with higher scores | **Association**. Possible other factor | Prior experience | May affect both. Missing **evidence** | Comparable starting conditions | Limits causal claim

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



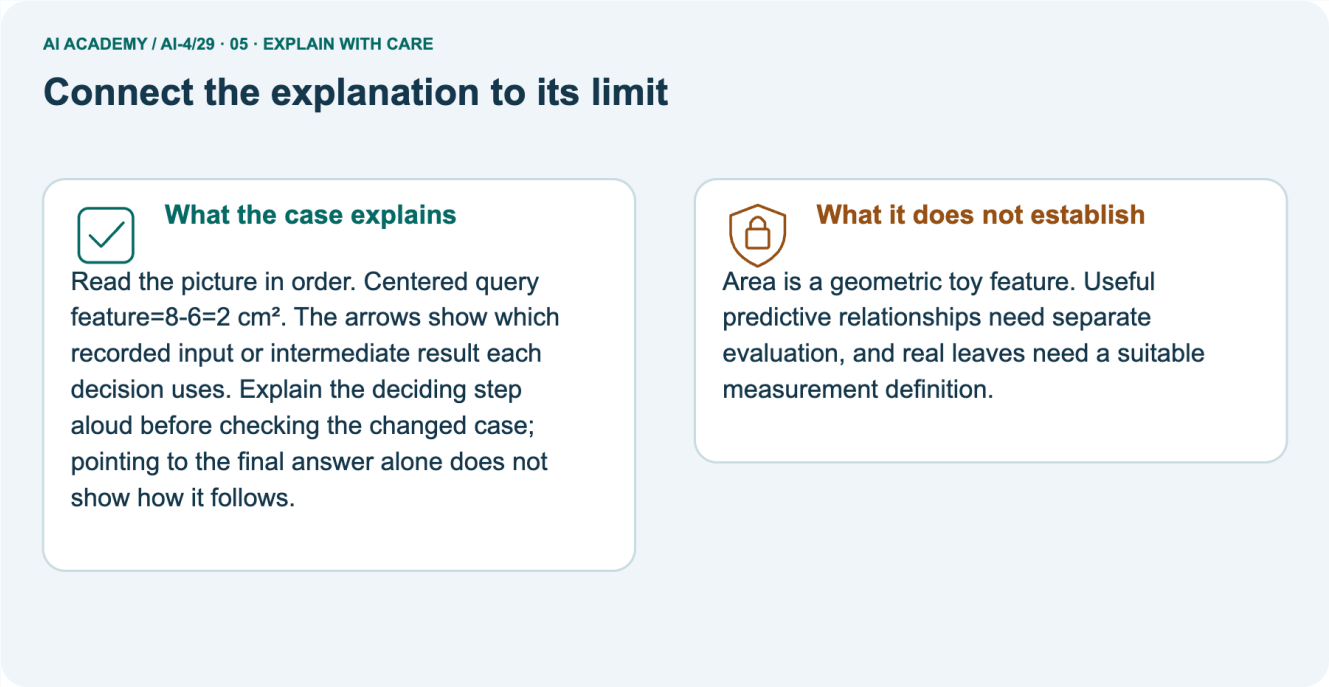
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Centered query feature=8-6=2 cm². Changed condition: Include the query in the centering mean. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

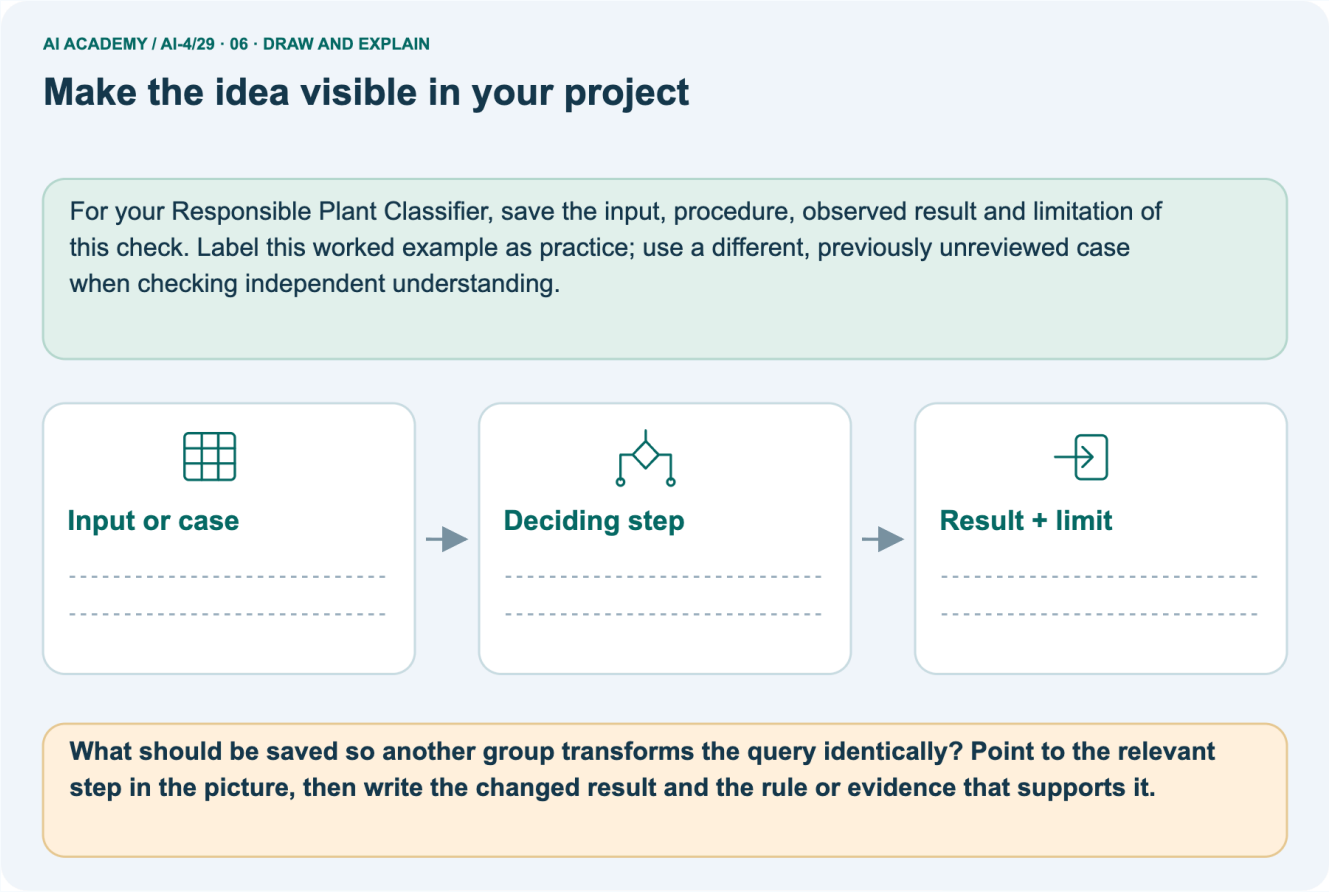
Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Area is a geometric toy feature.

Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Feature Engineering: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.